

TaxaExpect

Estimate spatially-explicit Bayesian priors for species occurrence. Part of the TaxaID ecosystem.

Bayes' Theorem improves taxonomic assignment by considering the prior probability that a hypothesized taxon occurs at the sampling location. TaxaExpect estimates these priors from occurrence records (from TaxaFetch or user-supplied data). Although occurrence data are often sparse and biased, they are usually sufficient to distinguish among taxa with similar match scores but very different geographic ranges.

TaxaExpect places occurrence data on a spatial grid (whose resolution is optimized automatically) and fits a hierarchical model that accounts for habitat type and spatial autocorrelation. These grid-based priors can be mapped over occurrences using `plot_theta_map_interactive()`, so that TaxaExpect becomes a full species distribution model. The model can also incorporate environmental covariates such as temperature or altitude, and it adjusts confidence based on sampling effort at each site. For taxa never reported in the study area, TaxaExpect generates “dark diversity” priors based on singleton observations elsewhere. Spatial priors reduce false positives from ecologically implausible assignments and can break ties between similar-scoring species, rescuing species-level resolution that would otherwise be lost to defensive upranking.

Overview

TaxaExpect generates theta priors (occupancy x detectability) for taxonomic assignment from occurrence data. Uses hierarchical spatial models to estimate expected species composition at grid cells, incorporating habitat type and geographic distance effects.

Three model tiers: - **Tier 1** – species observed at the site (direct estimate) - **Tier 2** – species observed nearby (spatial interpolation) - **Tier 3** – species expected but unobserved (dark diversity).

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaExpect")
```

Quick Start

```
library(TaxaExpect)

# Option A: End-to-end wrapper (fetches GBIF, assigns habitat, builds priors)
priors <- build_priors(
  taxa = species_df,          # data frame with taxonomy columns
  geometry = bbox_wkt,       # WKT bounding box from TaxaFetch
  main_habitat = "Marine"    # site habitat
)

# Option B: Step-by-step pipeline
# 1. Grid the occurrence data
sites <- create_sites_from_grid(occurrences, grid_size = 0.25)

# 2. Optimize grid resolution
grid_results <- optimize_grid_size(occurrences, n_covariates = 2)
```

```

# 3. Prepare model data (zero-fill, add covariates)
model_df <- prepare_model_dataframe(sites)

# 4. Fit the spatial model
model_fit <- train_biodiversity_model(
  model_df,
  formula = presence ~ lat_r + lon_r + (1 | taxon_name)
)

# 5. Generate priors (Tier 1 + 2 + 3)
priors <- generate_full_priors(model_fit)

```

Key Functions

High-level wrapper: - `build_priors()` – end-to-end: GBIF fetch -> habitat -> grid -> model -> priors

Gridding: - `create_sites_from_grid()` – generate spatial grid cells from occurrences - `optimize_grid_size()` – find optimal grid resolution for the study area

Modeling: - `prepare_model_dataframe()` – zero-fill and add spatial covariates - `compute_moran_basis()` – Moran eigenvector maps for spatial autocorrelation - `screen_spatial_formula()` – evaluate candidate model formulas - `train_biodiversity_model()` – fit hierarchical spatial model (glmmTMB)

Prior generation: - `generate_full_priors()` – predict priors at target sites (Tier 1 + 2) - `generate_undetected_diversity()` – dark diversity priors (Tier 3)

Diagnostics and reporting: - `plot_theta_map_interactive()` – interactive Leaflet map of priors - `report_priors()` – generate report section for `assemble_report()`

Statistical Methods

TaxaExpect fits a binomial generalized linear mixed model (GLMM) via glmmTMB (Brooks et al. 2017). The response is the count of species s at site g out of the total community count, modelled on the logit scale with:

- **Fixed habitat effects** capturing baseline differences across habitat types (Marine, Freshwater, Terrestrial, or IUCN L1 categories)
- **Random species intercepts** allowing each species its own baseline rarity, shrunk toward the global mean
- **Random habitat slopes** (screened for data sufficiency) giving each species its own habitat preference
- **Random spatial gradients** (latitude, longitude) per species, capturing geographic range trends
- **Species x grid deviations** capturing local departures from the spatial surface; the variance of this term provides the principled ceiling for prior concentration

Model predictions are back-transformed via the delta method and converted to Beta(alpha, beta) priors through moment-matching. A phi cap (from the grid-level variance) prevents astronomically tight priors near boundary theta values, while a phi floor (default 2) prevents modelled priors from becoming less informative than dark diversity fallbacks. For undetected species, singleton mirrors and a global floor prior Beta(1, N-1) provide Tier 3 coverage.

Moran eigenvector maps (Dray et al. 2006) capture fine-scale spatial autocorrelation beyond the latitude/longitude gradients and are included by default in `build_priors()` (`moran_k` = 5).

For the full statistical derivation, assumptions, and references, see `inst/TaxaExpect_supplemental_methods.md`.

Vignettes

- Building Priors – full workflow guide

Part of TaxaID

TaxaExpect receives habitat-annotated occurrence data from TaxaHabitat and produces spatially-explicit priors for TaxaAssign (posterior computation).

Ecosystem: TaxaFetch -> TaxaHabitat -> **TaxaExpect** -> TaxaAssign

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- glmmTMB (for hierarchical biodiversity models)
- TaxaTools, TaxaFetch, TaxaHabitat (for the `build_priors()` high-level wrapper; in Suggests)

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).